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Paper Code : PEC-CS702B/PEC-IT702C/PECDS702B Soft Computing

UPID : 007556

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) Which of the following search techniques has the capacity to overcome the problem of local optima?
A. Genetic algorithms
B. Simulated annealing
C. Both (A) and (B)
D. None of the above

(II) Which of the following statements is true about a linguistic variable?
A. A linguistic variable has linguistic values
B. The values of a linguistic variable are fuzzy sets
C. Both (A) and (B)
D. None of the above

(III) Which of the following is known as Fuzzy Inference?
A. Generalized Modus Ponens
B. Generalized Modus Tollens
C. Both (A) and (B)
D. None of the above

(IV) Which of the following competitive nets is based on the idea that patterns close to each other should be mapped to clusters with physical proximity?
A. MAXNET
B. Kohonen's self-Organizing Map (SOM)
C. Learning Vector Quantization (LVQ)
D. None of the above

(V) Which of the following is not true about McCulloch-Pitts neurons?
A. The interconnections are unidirectional
B. All excitatory interconnections have the same weight
C. All inhibitory interconnections have the same weight
D. The activation is bipolar

(VI) Pheromone quantity in ACO is ___ proportional to path selection.

(VII) Which of the following is the last step of a fuzzy inference process?
A. Fuzzification
B. Defuzzification
C. Either (A) or (B)
D. None of the above

(VIII) What is the difference between hard and soft computing?

(IX) The interconnections of a Hopfield network are
A. Unidirectional
B. Bidirectional
C. Both (A) and (B)
D. None of the above

(X) Which of the following GA operators do not help in exploring the various parts of the search space?
A. Selection
B. Crossover
C. Mutation
D. None of the above

(XI) While applying Hecht and Nielsen criteria for termination of the backpropagation learning process, which of the following sets of patterns is used to determine the termination condition?

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A. Training patterns
B. Training-testing patterns
C. Both (A) and (B)
D. None of the above

(xii) Let U = {a, b, c}, and P=0.5/a+0.5/c be a fuzzy set on U. Then which of the following is true?
A. P is normal
B. P is sub-normal
C. Both (A) and (B)
D. None of the above

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Explain simulated annealing in brief. [5]
3. What is perceptron? Compare the units and functionalities between a biological neuron and an artificial neuron. [5]
4. The Exclusive-OR function is not linearly separable and hence a single-layer perceptron cannot simulate it. Justify it. [5]
5. Explain crossover and mutation with examples. [5]
6. Explain Lambda Cuts for fuzzy sets. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Define perceptron learning rule. [3]
   (b) How the linear separability concept is implemented using perceptron network training? [4]
   (c) Implement AND function with Binary inputs and bipolar targets using perceptron training algorithm up to 2 epochs. [8]
8. (a) Explain Mamdani Fuzzy Model. [5]
   (b) Consider the fuzzy rule R : If service is good Then customer is satisfied. Related universes are service-rating = {a, b, c, d, e}, and satisfaction-grade = {1, 2, 3, 4, 5} where the service-ratings a, b, c, d, e are in descending order and the satisfaction-grades 1, 2, 3, 4, 5 are in the ascending order. The fuzzy sets good-service and satisfied are defined as follows: [5]

$$good\text{-}service = \frac{1.0}{a} + \frac{0.8}{b} + \frac{0.6}{c} + \frac{0.4}{d} + \frac{0.2}{e}$$

$$satisfied = \frac{0.2}{1} + \frac{0.4}{2} + \frac{0.6}{3} + \frac{0.8}{4} + \frac{1.0}{5}$$

   Find the relation matrix for this rule according to Jadeh's interpretation.
   (c) Discuss how Fuzzy Logic is applied in Weather forecasting. [5]
9. (a) Describe the backpropagation learning algorithm. [5]
   (b) With a training dataset, illustrate the working of the backpropagation learning algorithm. [5]
   (c) Explain GA based clustering algorithm. [5]
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10. (a) Write down the differences between Adaline and Madaline networks. [5]
   (b) Use Adaline network to train ANDNOT function with bipolar inputs and targets. Perform 2 epochs of training. [10]
11. (a) Write Short Notes on (Any two) [10]
   Tabu search
   Ant Colony Optimization (ACO)
   Particle Swarm Optimization (PSO)
   (b) Design a Hebb net to implement logical AND function with bipolar inputs and target. [5]

*** END OF PAPER ***

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